Vigil MD: Label and Feature Experiments

November 10, 2026 · Hitika Dalwadi · branch `model-dev/training-diagnostics` (commits `20a5dad`, `170c0c1`; E3/F3 not yet committed) · label sets built from the `vigil_md_prod` snapshot of 2026-11-07

Summary

The training diagnostics showed that the same-device model learned a shortcut: the match kind alone gave the label on 95.9% of training rows, because no training negative had agreeing identifiers and every row with a missing MAUDE model was a MATCH. On the hand-labeled sheet the baseline `xgb_all` accepted 42 of 83 NON_MATCH pairs and 45 of 48 INSUFFICIENT_EVIDENCE pairs (precision 0.628). We ran seven tracked experiments that change the labels, the features or the decision rule. **Hand-sheet precision rose from 0.628 to 0.959 for the best XGBoost model, with recall unchanged at 1.000.** Three changes did the work: hard negatives whose identifiers agree (E1), a TF-IDF text cosine feature (F1) and a rule that routes insufficient-evidence pairs away from the model (E4).

> **Recommendation:** use `label_set_v2` (E1 + E3), feature set `fs2` (F1 + F2) and the IE gate (E4). The best candidate is `xgb_no_match_kind`: precision 0.959 (3 false positives, 95% lower bound 0.899), recall 1.000. Do not pick between it and logistic regression yet: the hand labels follow the same rule tree the models learn, so these numbers measure agreement with the guide. A second, blind hand batch is the next step.

1. What was tested

Item	Value
Question	Does a label or feature change make the model reject wrong candidate devices on real, hand-labeled MAUDE reports without losing true matches?
Main evaluation	Hand sheet `hand_label_sample_v1`: 202 candidate pairs, of which 154 decided (71 MATCH, 83 NON_MATCH) and 48 INSUFFICIENT_EVIDENCE (IE)
Second evaluation	Held-out test split of each label set (about 125,000 UDI-linked pairs)
Models	Logistic regression and three XGBoost variants: all features, without the match kind, and without every kind-derived feature. Hyperparameters by random search (10 trials); the trial and the 99%-precision cutoff are picked on two separate validation halves
Metrics	Precision and recall on the decided hand pairs at the validation cutoff, with a 95% lower bound; IE pairs accepted (a wrong link even though precision leaves them out); diagnostics D1-D11 (shortcut purity, SHAP push on false positives)
Tracking	Every run in MLflow with tags `exp`, `git_commit`, label-set manifest hash, caps and `feature_set`; outputs per experiment in `data/runs/<exp>/`; registry in `docs/EXPERIMENTS.md`
Cost	BigQuery: about 1.1 GB processed for label_set_v2 (estimated first), 0.22 GB per sibling-key query. Training runs on a laptop in WSL (10 GB)

2. The experiments

ID	Change	Kind	Result
P0-rerun	Baseline, label_set_v1, 19 features (fs1), split validation	baseline	reference
E4	IE gate before the model: no usable MAUDE identifier (E6) or kind prefix_related (E3b) is IE, never MATCH	decision rule	**keep**
E1	Hard negatives with agreeing identifiers: same key at another labeler (no alias, other product code), contains hits at the same labeler, non-distinctive keys with a conflicting catalog (`label_set_v2_e1`)	labels	**keep**
F1	`feat_text_cosine`: TF-IDF character 3-5-gram cosine of the two record texts, encoder fitted on train-split GUDID texts only	feature	**keep**

ID	Change	Kind	Result
F2	`feat_contains_extra_len`, `feat_contains_at_end`	feature	keep (small effect)
E3	Placeholder-model negatives: missing MAUDE model with a usable catalog, paired with other models of the same labeler and product code (`label_set_v2` = E1 + E3)	labels	**keep** (see costs)
F3	`feat_labeler_sim`: trigram Jaccard of the labeler names (fs3)	feature	drop (no gain)

3. Results on the hand sheet

Precision on the 154 decided pairs at each model's validation cutoff. In brackets: false positives, then recall. '+ gate' rows apply the E4 gate at scoring. Each label or feature change builds on the one before; F3 is compared with the E3 row.

Experiment	xgb_all	xgb_no_match_kind	xgb_no_kind_derived	logreg_v1	IE accepted
P0-rerun (baseline)	0.628 (42, 1.00)	0.640 (40, 1.00)	0.676 (34, 1.00)	0.602 (47, 1.00)	45 / 48
E4: gate on P0	0.689 (32, 1.00)	0.676 (34, 1.00)	0.676 (34, 1.00)	0.689 (32, 1.00)	15 / 48
E1: agree negatives	0.835 (14, 1.00)	0.877 (10, 1.00)	0.826 (15, 1.00)	0.933 (5, 0.99)	44 / 48
F1+F2 on E1	0.899 (8, 1.00)	0.910 (7, 1.00)	0.922 (6, 1.00)	0.933 (5, 0.99)	44 / 48
F1+F2 + gate	0.934 (5, 1.00)	0.947 (4, 1.00)	0.922 (6, 1.00)	0.972 (2, 0.99)	14 / 48
E3 (label_set_v2) + gate	0.899 (8, 1.00)	**0.959 (3, 1.00)**	0.947 (4, 1.00)	1.000 (0, 0.94)	14 / 48
F3 on v2 + gate	0.899 (8, 1.00)	0.934 (5, 1.00)	0.947 (4, 1.00)	1.000 (0, 0.94)	14 / 48

Green: best model overall. Red: the two side effects of E3 (xgb_all loses precision, logistic regression loses recall). Without the gate, E3 gives xgb_no_match_kind 0.922 and xgb_all 0.866. 95% lower bounds: 0.899 for the best cell, about 0.82-0.88 for the other XGBoost cells of the last three rows.

4. Did the shortcut go away?

Diagnostics on the training data. A single-label kind means the match kind alone gives the label.

Label set	Rows with a single-label kind (D1)	contains: MATCH share (train)	missing model: MATCH share (train)	Test-split recall (xgb_all)
label_set_v1 (P0)	95.9%	100%	100%	0.999
label_set_v2_e1 (E1)	52.9%	24.3% (1,357 of 5,590)	100%	0.994
label_set_v2 (E1 + E3)	49.1%	24.3%	39.4% (11,631 of 29,555)	0.965

Test-split precision stays at 0.99 throughout. Test-split recall falls in label_set_v2 because rows with a missing model are now genuinely ambiguous: the same pattern is a MATCH in some rows and a NON_MATCH in others.

5. Findings

1. **The labels were the problem, not the model family.** Before E1 the rule 'MATCH if the kind was always MATCH in train' had test precision 1.000, so no model could do better on the label set while failing on the hand sheet. Adding negatives whose identifiers agree (E1) cut hand false positives of xgb_all from 42 to 14.
2. **Text similarity separates substring collisions.** The cosine is a top-2 feature by SHAP in every model. The 9 contains false positives (for example MAUDE model 1260 inside Boston Scientific catalog M001711260, device names MACH1 vs WALLSTENT) have cosines of 0.09-0.38, while true matches of that kind have a median of 0.73; xgb_all's contains false positives fell from 9 to 3.
3. **Text cannot tell suffix siblings apart.** GP200 vs GP200S style pairs have cosine 0.81-0.89. The IE gate is the right tool there: it moved IE pairs accepted from 45 to 14 of 48 without losing a true match.

4. **Placeholder negatives fixed the 'missing model means MATCH' rule** (E3). The SHAP push of feat_maude_model_missing on false positives turned from +0.6..+2.7 to -0.5..-1.1, and both placeholder false positives (catalog 71501-01 vs models 70817 / 70819) are gone.
5. **E3 has costs.** Logistic regression now penalizes a labeler-name mismatch and misses 4 true matches (the same key at labelers with different names, such as 'sistema de implante nacional' vs 's.i.n. implant system'). xgb_all lets more contains pairs through (3 to 8 false positives, from a few repeated keys).
6. **Labeler-name similarity (F3) did not help.** No model improved and the target miss stayed missed; the feature set fs3 stays defined so the run can be reproduced, the default is fs2.

6. What is still wrong

False positives of the recommended model and of xgb_all on label_set_v2 with the gate:

- **contains, short numeric keys** (3401, 8101, 1260 inside long catalog codes of the same labeler and product code): 3 for xgb_no_match_kind, 8 for xgb_all. Same labeler, same product code and a substring hit are strong evidence to the model; only the text cosine argues against them.
- **prefix_related** (BLX500H15 vs BLX500H15MQ, MDL2513 vs MDL2513SZ): removed by the gate, but still accepted by the models themselves.
- **INSUFFICIENT_EVIDENCE**: 14 of 48 IE pairs are still accepted after the gate; they have agreeing identifiers but the guide asks for more evidence (non-distinctive keys, reverse suffix cases).

7. Limits

- **The hand labels follow the rule tree** (diagnostic D8: 154 of 154 decided pairs agree with it). Precision on this sheet measures agreement with the labeling guide, not independent ground truth.
- **The sheet is small.** 83 NON_MATCH pairs, and several false positives repeat the same key. A difference of 2-3 false positives is within noise; the 95% lower bound of the best result is 0.899.
- **One run per experiment**, 10 random-search trials each. F1 and F2 were run together, so F2 alone is not measured.
- **Model selection still uses the label-set validation split**, which is easier than the hand sheet (validation AP 0.998).

8. Recommendations and next steps

1. **Adopt label_set_v2 + fs2 + IE gate** as the new baseline. Put the gate into the scoring path, not only into the evaluation script.
2. **Label a second hand batch blind** (no helper columns), stratified by manufacturer size, renamed labelers, missing identifiers and contains cases. Use it to choose the model and the cutoff; keep batch 1 as the test set.
3. **Target the contains collisions**: hard negatives drawn from the TF-IDF neighbours of each anchor (E2, needs the embedding export), or a feature for how many GUDID records of the labeler contain the same key.
4. **Narrative identifiers** (F4): match candidate model and catalog numbers against the MAUDE narrative for rows without a usable identifier, after a cost estimate of the maude_text query.
5. Then reweight training rows by match kind and add monotone constraints (F5), once the hand batch can show their effect.

9. How to reproduce

```
cd Model-Development/matching
# label set v2 (E1 + E3): estimate first, then build
python scripts/s01_build_label_set.py --project $P --only 04c 05 06 --label-set label_set_v2 \
    --agree-neg-per-anchor 2 --placeholder-neg-per-anchor 2 --easy-anchor-pct 25 --estimate-only
python scripts/s01_build_label_set.py ... --snapshot-date 2026-11-07
python scripts/s03_export_label_set.py --project $P --table label_set_v2 --out data/label_set_v2.parquet
python scripts/s04_build_feature_table.py --project $P --table label_set_v2 --data data/label_set_v2.parquet \
    --manifest labels/label_set_v2.manifest.json --out data/label_set_v2_fs2_features.parquet \
    --siblings data/sibling_keys_v2.csv --feature-set fs2
python scripts/s05_build_hand_features.py --feature-set fs2 \
    --tfidf data/label_set_v2_fs2_features.tfidf.pkl --out data/hand_features_v2_fs2.csv
export VIGIL_EXP=E3-placeholder-neg
python scripts/s06_train_models.py --data data/label_set_v2_fs2_features.parquet \
    --manifest labels/label_set_v2.manifest.json --hand data/hand_features_v2_fs2.csv
python eval/s08_all.py && python eval/s08_all.py --ie-gate
python scripts/diag_training_issues.py --features data/label_set_v2_fs2_features.parquet \
    --label-set data/label_set_v2.parquet --manifest labels/label_set_v2.manifest.json \
    --siblings data/sibling_keys_v2.csv --hand-features data/hand_features_v2_fs2.csv
```

Outputs: `data/runs/<exp>/` (scores, cutoffs), `reports/training_diag/<exp>/` (RESULTS.md, SHAP tables), MLflow runs tagged `exp`. All results per experiment: `docs/EXPERIMENTS.md`. Note: the local MLflow store was lost on 2026-11-09 when the Docker MLflow server and the scripts wrote it at the same time; P0, E4 and E1 were re-logged. Keep a single writer (the scripts, with `mlflow ui` for viewing).